Brian, Toan, and Montserrat (TBAMM)

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Reflection Journal

Brian: Introduction and Workflow Understanding

In Lab 07, the focus was on understanding how to evaluate a machine learning model and why accuracy alone is not enough to find out whether a model performs well. The purpose of this lab was to create the Titanic dataset to train a logistic regression model and to examine its predictions using different techniques. At first, the workflow did seem like different steps that included data preparation, model training, and evaluation. However, as I worked through the notebook, I realized that each step is connected to the next. I learned that preparing data is important because the model depends on the features provided to it. In this lab, I worked with features such as age, passenger class, sex, and fare to predict whether a passenger survived. This helped me understand how machine learning uses information from a dataset to find patterns and make predictions.

One important thing I learned was that training a model is only one part of the machine learning process. After training the logistic regression model, I needed to even examine its predictions and find how well that is matched with the actual outcome. I also learned why the dataset is split into training and testing sets, since evaluating a model on separate test data helps show how the model will perform on examples it was not trained on. Working through the code

and reviewing the outputs helped me connect the programming steps with their purpose. I realized that a model can make predictions without being accurate or reliable in every scenario. This changed how I thought about machine learning because I now understand that building a model is not just about getting it to produce an answer; instead, it is about checking performance, identifying possible errors, and understanding what those results actually mean. It helped build a foundation for learning about the confusion matrix, precision, recall, and F1-score, which gave me more detail on how a model performs.

Toan: Evaluation Metrics and Critical Thinking

One important thing I learned in Lab 07 was how to use a confusion matrix to understand a machine learning model's predictions. A confusion matrix shows four types of results: true positives, true negatives, false positives, and false negatives. True positives are cases where the model correctly predicts a positive result, while true negatives are cases where it correctly predicts a negative result. False positives happen when the model predicts a positive result incorrectly, and false negatives happen when the model misses an actual positive case. I also learned that precision and recall measure different aspects of a model's performance. Precision tells us how many of the model's positive predictions are actually correct, while recall measures how many of the actual positive cases the model successfully identifies. The F1-score combines precision and recall into one measurement, which is useful when we want to consider both at the same time. These metrics helped me understand that accuracy alone does not provide enough information to judge whether a model performs well.

I also learned that choosing the right evaluation metric depends on the problem and the consequences of making incorrect predictions. For example, in an email spam filter, a false

positive could cause an important email to be incorrectly marked as spam. In medical applications, a false negative could mean that a serious condition is missed, which may lead to delayed treatment. These examples helped me understand why different projects may need to focus on different metrics. If a model needs to identify as many positive cases as possible, recall may be especially important. However, if incorrect positive predictions create serious problems, precision may deserve more attention. By examining the confusion matrix and individual types of errors, I can better understand the model's weaknesses and determine where it needs improvement. Overall, I learned that evaluating a machine learning model requires more critical thinking than simply checking its accuracy score because the types of mistakes it makes can be just as important as its overall performance.

Montserrat: Cross-Validation and Future Improvements

In this lab, I learned how five-fold cross-validation can help evaluate a machine learning model more reliably. This method divides the dataset into five parts, using four parts to train the model and the remaining part to test it. This process is repeated five times so that each part is used for testing once. According to the lab report, the average cross-validation score was 78.68%, with a standard deviation of 0.0094. I understand that the average score represents the model's performance across the five folds, while the standard deviation indicates how much the results varied. The relatively small standard deviation suggests that the model performed consistently across different data splits. I learned that cross-validation provides a more reliable estimate of model performance than relying on a single train-test split because the results are less

dependent on one division of the data. This helped me understand why evaluating a model in multiple ways is important before deciding whether it performs well.

This lab improved my understanding of how to evaluate machine learning models more carefully instead of focusing on only one accuracy score. I learned that cross-validation, average scores, and standard deviation provide useful information about a model's performance and consistency. In future machine learning projects, I can apply these methods to compare different models and better understand how well they may perform on unseen data. Next time, I would like to compare multiple algorithms, examine incorrect predictions to understand the model's weaknesses, and make sure the evaluation method and performance metrics are appropriate for the problem. For example, some projects may require more attention to precision and recall rather than accuracy alone.

In conclusion, this lab changed my understanding of machine learning model performance by showing me that a good model is not simply one that produces a high score. It is also important to determine whether its results are consistent, reliable, and suitable for the task it is designed to solve. I now understand that evaluating a model requires more than checking its initial results. Using methods such as five-fold cross-validation can provide a clearer picture of how a model performs across different data splits. I will use what I learned in future projects to make better decisions when training, testing, and comparing machine learning models.